tion. This flexibility ensures user-friendly technology utilisation. Fig. 3 illustrates the implantable RFID chip and Fig. 4 demonstrates the chip's implantation process using a syringe.

Whether you use a vet or are making it for learning, the code and working process can proceed without actual implantation. For testing purposes, animal like rats, snakes, or fish can be used. However, for real-life deployment inside the body for health and data tracking, careful consideration is needed. Refer to the following video tutorials for detailed implantation processes:

https://www.youtube.com/watch?v=GsObVs8QuWE

https://www.youtube.com/watch?v=zotc4sKyWRA.

Software coding

After implanting the chip, set up the reader device. This device scans the implanted RFID, cross-references its data with the database, and prompts for a unique name assignment if the data is new. To initiate this process, install the necessary libraries using the following commands:

```
sudo pip3 install pyserial
sudo pip3 install rdm6300
```

Next, develop code that imports the required library, sets the serial port name, and scans the CSV or sheet file to locate the RFID number. Retrieve the stored animal name linked to that RFID number, facilitating animal identification (see Fig. 6).

Moving forward, establish a while loop to implement RFID checking. Continuously monitor RFID scans, extract unique codes, check for existing data in the database, display the animal's name if present, or prompt the user to provide a name for a new RFID. Save this new name and RFID data in a CSV file. Fig. 7 shows the loop snippet.

Fig. 8: Chip scanner attached with an antenna to read the chip data

+ Sheet 1

rfid_data

RFID	Name	Date	Time
23020061	Snake236 Cobra	2023-08-23	18:02:49
23020061	SNAKE 234 Cobra	2023-08-23	18:04:01
23020061	SNAKE 234 Cobra	2023-08-23	18:04:11

Fig. 9: Animal identification names stored in the database

```
Please insert an RFID card
Enter a name for the RFID 23020061: SNAKE 234 Cobra
RFID 23020061 saved with name SNAKE 234 Cobra on 2023-08-23 at 18:04:01
Enter a name for the RFID 23020061: SNAKE 234 Cobra
RFID 23020061 saved with name SNAKE 234 Cobra on 2023-08-23 at 18:04:11
Enter a name for the RFID 23020061: █
```

Fig. 10: Code detecting the RFID and asking to enter the unique identification name for that animal

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

Fig. 11: The chip used in this device

Construction and testing

Connect the components according to the circuit diagram in Fig. 2. Connect FTDI to RDM6300 RFID reader and link it to your laptop. Execute the Python code that has been developed. When you scan an animal, the system performs actions like scanning the RFID code, checking the database for registration, prompting for a unique name if it's a new RFID, and saving the name in the database. Fig. 8 illustrates the chip scanner attached with an antenna, and Fig. 9 shows the animal identification names stored in the database.

This process effectively utilises RFID technology for accurate and comprehensive animal tracking. Fig. 10 shows the code detecting the RFID and prompting for a unique identification name for that animal.

EFY note. This marks the first part of a series of articles on Implantable Electronics. Stay tuned for the next version involving IoT data tracking, storing data inside the body, and real-time position tracking system using implantable electronics. Keep checking Electronics For You for the next part. EFY

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